

Long-Term Solvency & Liquidity Ratios - Level 5/6 Student Resource

Understanding Long-Term Solvency & Key Ratios

Long-term solvency describes an organisation's ability to meet long-term obligations. It focuses on financing structure (debt vs equity) and the ability to service interest payments.

1. Debt / Equity Ratio (Gearing Ratio)

Formula: $\text{Debt} / \text{Equity} = \text{Total Long-Term Debt} / \text{Total Equity}$

Shows the proportion of debt relative to shareholders' funds. High gearing = higher financial risk.

2. Debt / (Debt + Equity) Ratio

Formula: $\text{Debt} / (\text{Debt} + \text{Equity}) = \text{Total Long-Term Debt} / (\text{Total Debt} + \text{Total Equity})$

Shows the percentage of total capital funded by debt.

Differences Between Ratios

Debt/Equity is a multiple; Debt/(Debt+Equity) is a proportion. The first compares debt to equity only; the second compares debt to total capital.

3. Interest Cover (PBIT / Finance Charges)

Formula: $\text{Interest Cover} = \text{PBIT} / \text{Finance Charges}$

Shows how many times a company can cover its interest payments.

Other Solvency Ratios

- Long-Term Debt Ratio = $\text{Long-Term Debt} / \text{Total Assets}$
- Cash Flow to Debt = $\text{Operating Cash Flow} / \text{Total Debt}$

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Worked Examples (Three-Year Comparison)

Fictitious company: AlphaTech Ltd.

Financial Data

2023: Debt 40, Equity 60, PBIT 25, Finance Charges 5

2024: Debt 55, Equity 65, PBIT 22, Finance Charges 7

2025: Debt 70, Equity 70, PBIT 20, Finance Charges 10

Debt / Equity Calculations

2023: $40/60 = 0.67$

2024: $55/65 = 0.85$

2025: $70/70 = 1.00$

Debt / (Debt + Equity) Calculations

2023: 40%

2024: 46%

2025: 50%

Interest Cover Calculations

2023: $25/5 = 5.0$

2024: $22/7 = 3.1$

2025: $20/10 = 2.0$

Summary Table

Debt/Equity: 0.67, 0.85, 1.00

Debt/(Debt+Equity): 40%, 46%, 50%

Interest Cover: 5.0, 3.1, 2.0

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Overall Evaluation

AlphaTech is becoming more highly geared, with rising debt levels and falling interest cover, indicating increased solvency risk.